

→ WAVE MOTION ←

Wave = wave is a disturbance in a medium in which energy is transferred, without matter, from one place to another place.

→ Necessary properties of medium for wave transmission

- ① Resistance should be negligible.
- ② It should have property of Inertia - opposing nature.
- ③ It be elastic in nature - Elasticity

particles vibrate around mean position

→ In wave motion

- ① Energy = Transfer
- ② matter = not Transfer

Types of waves =

① On the basis of medium

(a) mechanical wave or Elastic waves =

These waves which required medium to travel

Ex = Sound wave

(b) Electro-magnetic wave =

In these waves medium is not required to travel

Ex = light

Mechanical wave = Two types

(a) Longitudinal waves = In

These waves the particles of medium vibrate in the direction of wave transmission, and the transmission is in the form of compression and expansion.

Example = Sound waves

Transmitted in air, waves transmitted inside water etc.

(b) Transverse waves = In

These waves the particles of the medium vibrate in the direction perpendicular to the direction of wave transmission.

Example = waves transmitted in stretched threads,

waves transmitted on the surface of water etc.

→ The speed of transverse and longitudinal waves in the same medium is different.

Based on energy transmission =

(a) progressive waves =

The waves that move ahead in the medium with a definite velocity and by which energy is transmitted are called progressive waves.

The matter through which the wave is transmitted does not move.

(b) Stationary waves =

The waves that do not move ahead in the medium but remain steady between the two limits of the medium are called stationary waves.

Based on the dimensions =

(a) One dimensional wave

In these the energy is transferred only along

one axis or direction like wave in stretched thread.

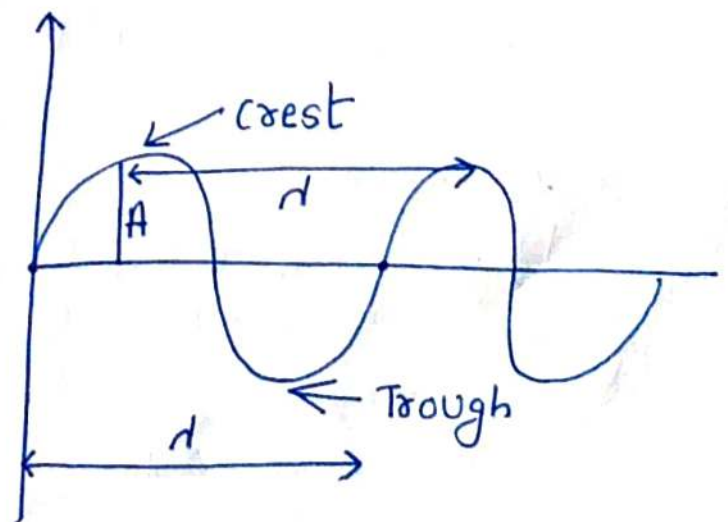
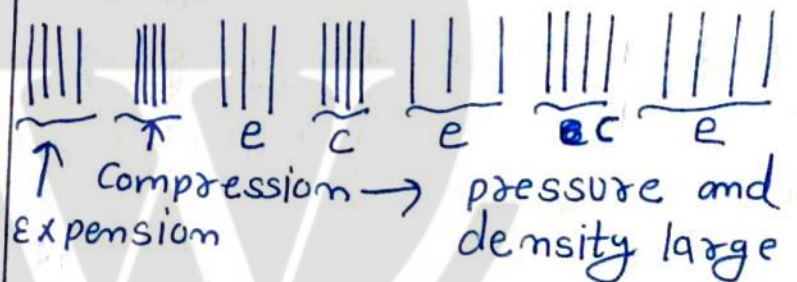
(b) Two dimensional waves =

In these the energy is transferred in a plane like waves produced on throwing a pebble on the surface of water.

(c) Three dimensional waves =

In these energy is transferred in the free space like sound and light waves transmitted in the space.

Special about Longitudinal =

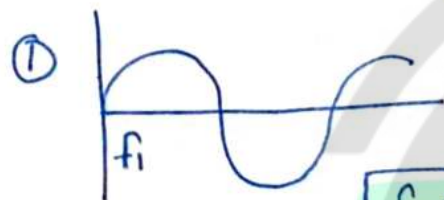


→ maximum displacement from mean position
(Amplitude)

→ wavelength (λ) =

- ① length of continuous crest and trough
- ② Distance B/w two continuous crest or trough

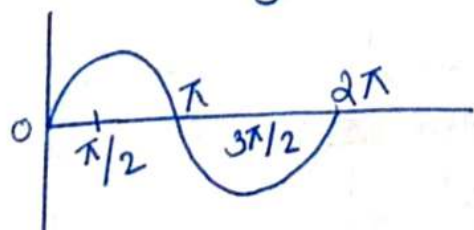
→ frequency (f) no. of cycle per sec = $f = \frac{1}{T}$



Definitions related to wave motion =

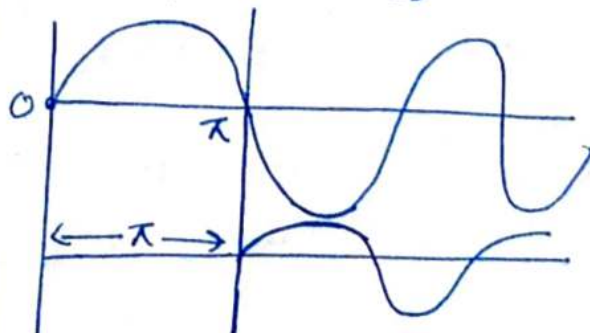
① phase or phase angle =

The physical quantity which expresses the position of motion of the vibrating particle from the equilibrium position is called phase or phase angle.



② phase difference = The difference

In the phase angles of two vibrating particles is called phase difference.



③ wave number = The number of waves present per unit length is called wave number.

$$n = \frac{1}{\lambda}$$

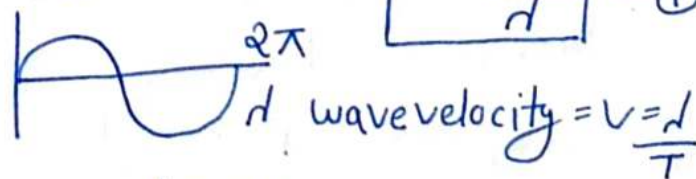
④ wave velocity = The distance covered by a wave in a definite direction in one second is called wave velocity.

$$v = \frac{\lambda}{T}$$

⑤ propagation constant (k) =

The phase difference between the particles situated at unit distance is called propagation constant or angular wave number.

$$k = \frac{2\pi}{\lambda} \quad \text{--- (1)}$$



$$\lambda = vT$$

from equation (1)

$$k = \frac{2\pi}{vT} \quad \left[f = \frac{1}{T} \right]$$

$$k = \frac{2\pi f}{v}$$

$$k = \frac{\omega}{v}$$

$$[\omega = 2\pi f]$$

Sound waves = The waves due to the vibrations of which our ear drum vibrates and the vibration signals are propagated to the brain and give us the impression of sound are called sound waves.

→ Audible Sound = $20\text{Hz} < f < 20\text{kHz}$

→ Ultrasonic Sound = $f > 20\text{kHz}$

→ Infrasonic Sound = $f < 20\text{Hz}$

Shock = when velocity is more than sound wave

properties of sound waves =

① Medium is required for their propagation.

② These waves need time to move from one place to other and their velocity at normal temperature and pressure is 332m/s

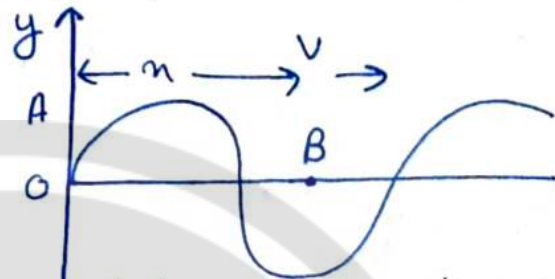
③ These are longitudinal in nature.

④ The sound waves show reflection, refraction,

Interference and diffraction like light waves.

⑤ The sound waves do not polarize because they are longitudinal in nature.

⇒ Displacement Equation of a progressive wave =



Let a progressive wave is moving with Amplitude (A) then displacement equation

$$y = A \sin \omega t \quad \text{--- ①}$$

→ At point (B) the displacement is cover in time $t = \frac{x}{v}$

$$\text{time} \Rightarrow \left[t - \frac{x}{v} \right]$$

$$y = A \sin \omega \left[t - \frac{x}{v} \right]$$

$$y = A \sin \frac{2\pi}{T} \left[t - \frac{x}{v} \right] \quad \left[\omega = \frac{2\pi}{T} \right]$$

$$y = A \sin \left[t \frac{2\pi}{T} - \frac{x}{v} \frac{2\pi}{T} \right]$$

$$y = A \sin \left[t \frac{2\pi}{T} - \frac{2\pi x}{\lambda} \right] \quad [\lambda = vT]$$

$$y = A \sin [t\omega - xk]$$

$$y = A \sin [\omega t - kx]$$

$$\omega = \frac{2\pi}{T}$$

$$k = \frac{2\pi}{\lambda}$$

→ This is equation of displacement of progressive wave in +x direction.

→ Displacement equation in -n direction
 $y = A \sin(\omega t + Kn)$

phase and phase difference

Let A wave is in +n direction and displacement is n_1 at any time t , then displacement equation

$$y_1 = A \sin(\omega t - Kn_1) \text{ --- (1)}$$

and displacement equation of 2nd wave

$$y_2 = A \sin(\omega t - Kn_2)$$

→ phase angle

$$\phi_1 = \omega t - Kn_1$$

$$\phi_2 = \omega t - Kn_2$$

phase difference =

$$\Delta\phi = \phi_2 - \phi_1$$

$$\Delta\phi = (\omega t - Kn_2) - (\omega t - Kn_1)$$

$$\Delta\phi = -Kn_2 + Kn_1$$

$$\Delta\phi = K(n_1 - n_2)$$

$$\Delta\phi = K\Delta n$$

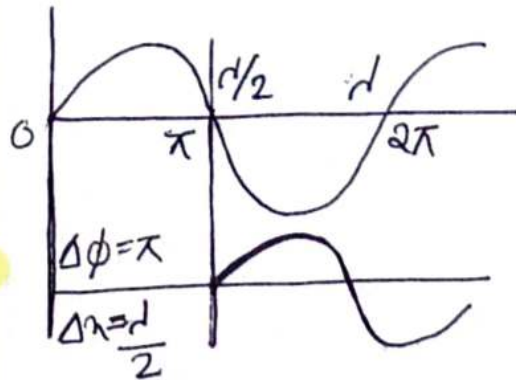
$$\Delta\phi = K\Delta n \quad \left| \quad K = \frac{2\pi}{\lambda} \right|$$

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta n$$

$$\frac{\Delta\phi}{2\pi} = \frac{\Delta n}{\lambda}$$

$$\Delta\phi = \text{phase d/f}$$

$$\Delta n = \text{path d/f}$$



Example = If the equation of the wave is $y = 5 \sin 5\pi \left(\frac{1}{0.02} - \frac{n}{20} \right)$ m then what would be its frequency, amplitude and time period?

Ans = Amplitude $A = 5$
 Time $T = 0.02$ Sec
 $f = \frac{1}{\text{Time}}$
 $= \frac{1}{0.02} = 50 \text{ Hz}$

Speed of transverse wave in a stretched string =

The speed of transverse wave in stretched string

$$V = \sqrt{\frac{T}{m}}$$

Does not depend upon the frequency of the wave.

T = Tension in string

m = mass per unit length

Speed of transverse wave =

$$V = \sqrt{\frac{T}{\mu}} \text{ or } \sqrt{\frac{T}{m}}$$

μ = mass per unit length

Velocity of Sound wave (Longitudinal)

Velocity depends on

- ① Elasticity $\uparrow\uparrow \rightarrow$ speed $\uparrow\uparrow$
- ② Inertia $\uparrow\uparrow \rightarrow$ speed $\downarrow\downarrow$

Newton's formula of

Sound wave = According to Newton the coefficient of elasticity is E and density of medium is d , then speed

$$v = \sqrt{\frac{E}{d}}$$

① Speed of Sound wave in Solid = According to Newton
Speed $v = \sqrt{\frac{E}{d}}$

If Y is young modulus of elasticity then speed is

$$v = \sqrt{\frac{Y}{d}}$$

② Speed of Sound wave in liquid = for liquid
Let B is bulk modulus of elasticity

$$v = \sqrt{\frac{B}{d}}$$

$\Rightarrow$ Speed in water = 1480 m/s at 20°C

③ Speed of Sound in gas

Ideal gas equation

$$\Rightarrow PV = nRT$$

for Constant Temperature

$$PV = \text{Constant}$$

$\rightarrow$ Differentiate the above Equation

$$Pdv + vdp = 0$$

$$Pdv = -vdp$$

$$P = -\frac{vdp}{dv} \quad \text{--- (I)}$$

Bulk modulus of Elasticity

$$B = -\frac{dP}{dv/v}$$

$$B = -v \frac{dP}{dv} \quad \text{--- (II)}$$

from (I) and (II)

Bulk = pressure modulus

$$B = P$$

According to Newton

$$v = \sqrt{\frac{B}{d}}$$

So
$$v = \sqrt{\frac{P}{d}}$$

At N.T.P.

$$P = 1.01 \times 10^5 \text{ pascal}$$

$$d = 1.3 \text{ kg/m}^3$$

$$\text{Speed } v = \sqrt{\frac{1.01 \times 10^5}{1.3}}$$

$$v = 279 \text{ m/s}$$

$\rightarrow$ But experimentally the speed is 332 m/s

Laplace Correction = According to Newton the transmission of sound wave is Isothermal process, But according to

to Laplace it is An adiabatic process.

→ for Adiabatic process

$$pV^\gamma = \text{constant}$$

→ Differentiate both side

$$p\gamma V^{\gamma-1}dv + V^\gamma dp = 0$$

$$p\gamma V^{\gamma-1}dv = -V^\gamma dp$$

$$p\gamma = \frac{-V^\gamma dp}{V^{\gamma-1}dv}$$

$$p\gamma = -\frac{dp}{V^{\gamma-1}V^{-\gamma}dv}$$

$$p\gamma = -\frac{dp}{V^{-1}dv}$$

$$p\gamma = -V \frac{dp}{dv} \quad \text{--- (I)}$$

Bulk modulus

$$\beta = -V \frac{dp}{dv} \quad \text{--- (II)}$$

from (I) and (II)

$$\beta = p\gamma$$

$$\text{Speed } v = \sqrt{\frac{\beta}{\rho}}$$

$$v = \sqrt{\frac{p\gamma}{\rho}} \quad \text{--- (I)}$$

$$\gamma = \frac{C_p}{C_v} \quad \left| \begin{array}{l} \gamma = 1.41 \text{ for air} \\ p = 1.01 \times 10^5 \text{ pascal} \\ \rho = 1.8 \text{ kg/m}^3 \end{array} \right.$$

$$v = \sqrt{\frac{1.41 \times 1.01 \times 10^6}{1.3}}$$

$$v = 331.3 \text{ m/s}$$

$$\boxed{v \approx 332 \text{ m/s}}$$

→ Speed of sound maximum in Solids minimum in gas

$$v = \sqrt{\frac{\beta}{\rho}}$$

$$\boxed{\beta_{\text{solid}} > \beta_{\text{gas}}}$$

factors affecting sound =

(I) Effect of Temperature =

Ideal gas equation

$$pV = nRT$$

$$\text{density} = \frac{\text{mass}}{\text{volume}}$$

$$V = \frac{m}{\rho}$$

$$p \frac{m}{\rho} = nRT$$

$$\frac{p}{\rho} = \left(\frac{nR}{m} \right) T$$

$$\text{Speed} = v = \sqrt{\gamma \frac{p}{\rho}}$$

$$v = \sqrt{\gamma \frac{nR}{m} T}$$

$$\boxed{v \propto \sqrt{T}}$$

Temp ↑ ↑ → speed ↑ ↑

(II) Effect of pressure =

$$\frac{p}{\rho} = \frac{RT}{m}$$

at constant Temperature, mass
constant $\frac{p}{\rho} = \text{constant}$

$$\boxed{v = \text{constant}}$$

→ Speed doesn't depend on p

(III) Effect of Humidity =

Humid = water + dry
air vapour air

→ Humidity ↑ ↑ → Density ↑ ↑ → Speed ↑ ↑

principle of Superposition

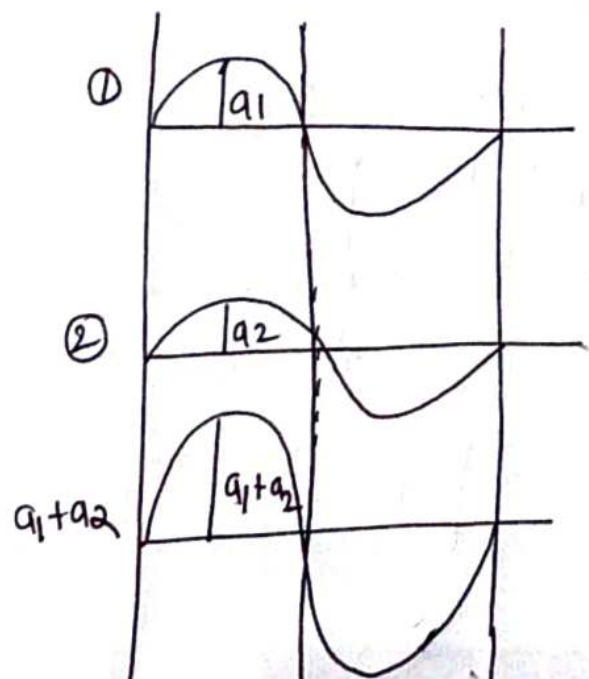
When two or more than two waves are superimposed, while propagating in medium, then Resultant displacement in that medium is equal to the sum of displacement of individual particle

→ Let the displacement of particle are $\bar{y}_1, \bar{y}_2, \dots$ and $\bar{y}_n$ then according to principle of superposition net displacement

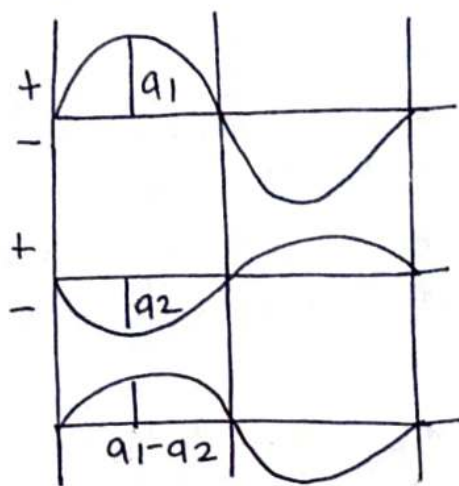
$$\bar{y} = \bar{y}_1 + \bar{y}_2 + \dots + \bar{y}_n$$

→ Example =

Let two waves of Amplitude A_1 and A_2 are superimpose in same phase.



⇒ If two waves are in opposite phase



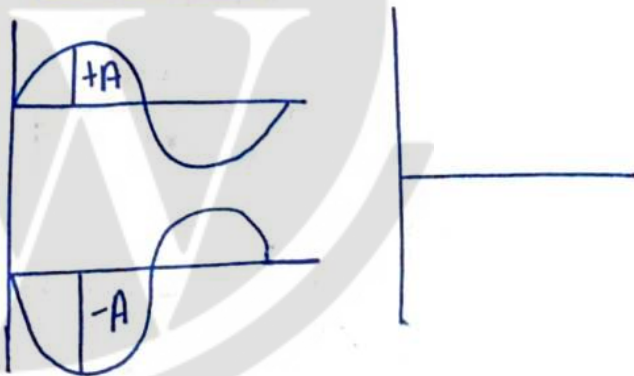
⇒ If $A_1 = A_2$ and waves are in opposite phase

→ Resultant amplitude

$$A_{net} = A_1 - A_2$$

$$A_{net} = A - A$$

$$A_{net} = 0$$

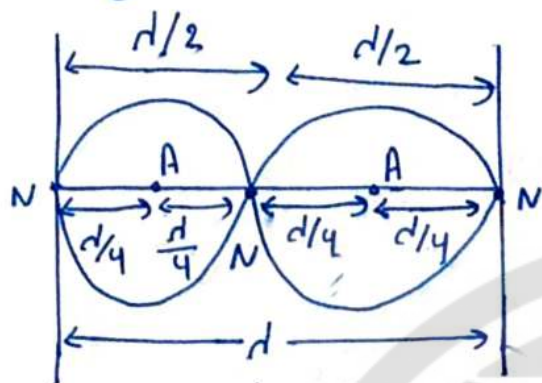


Stationary waves = when

Two waves of same frequency and same Amplitude are travelling in opposite phase with same velocity then according to principle of superposition Resultant wave does not progress.

① Energy Transfer is not possible but energy Re-distribution is possible

② In Longitudinal no change in pres



N = Nodes [Amplitude zero]

A = Antinodes [Amplitude maximum]

Nodes = The point of vibration at which Amplitude is Zero, called Nodes.

Antinodes = The point of vibration at which amplitude is maximum, called Antinodes

The distance B/w node to node = $\frac{\lambda}{2}$

The distance B/w Antinode to antinode = $\frac{\lambda}{2}$

The distance B/w 'N' and 'A' $\Rightarrow \frac{\lambda}{4}$

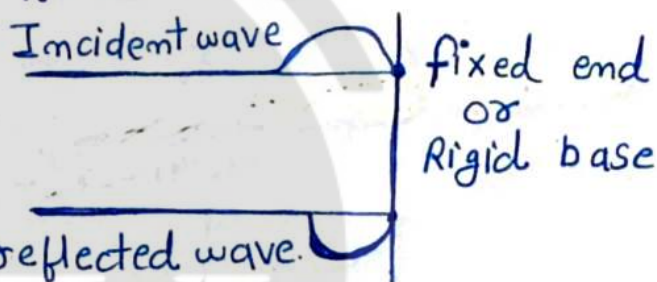
Nodes = $0, \frac{\lambda}{2}, \lambda, \frac{3\lambda}{2}, 2\lambda$

Antinodes = $\frac{\lambda}{4}, \frac{3\lambda}{4}, \frac{5\lambda}{4}$

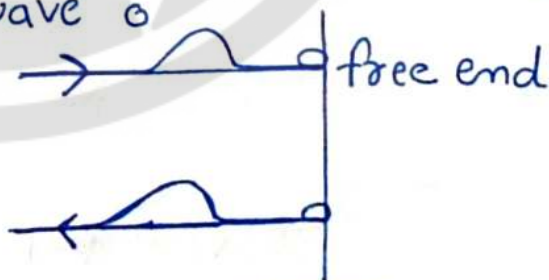
Reflection of wave = When a wave Bounce Back after striking than this phenomena is called reflection

Reflection By fixed end =

The phase difference in Incident and reflected wave is 180° or π rad.



Reflection of wave at free end = The phase difference B/w Incident and reflected wave is 0



Standing waves = When two Identical waves travelling in opposite directions in the same medium combine and form a steady wave pattern, this pattern is known as standing waves.

Note = Waves Combine

According to Superposition principle.

Some definitions =

⇒ fundamental frequency
or first harmonic =

"The minimum frequency or vibration of an object is called fundamental frequency."

⇒ organ pipe = The pipe in which air is filled, and by vibrating molecules sound is produced

Types of organ pipe =

① open organ pipe = The pipe at which both ends are open.

Note = ① at open end = Antinodes

② at close end = nodes

② Close organ pipe =

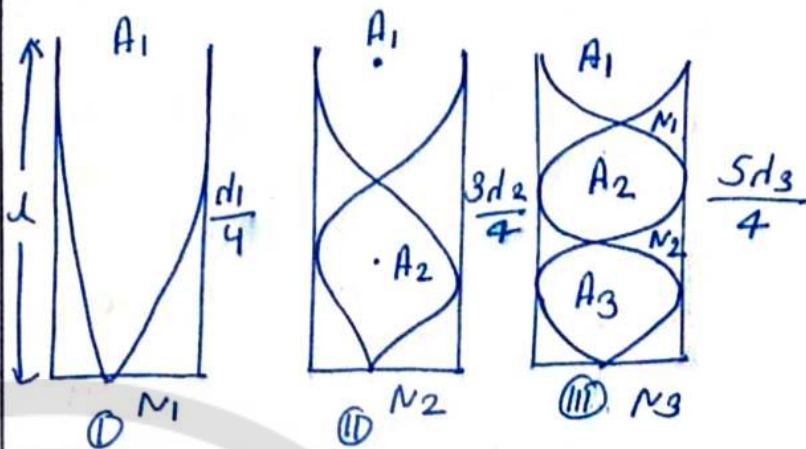
The organ pipe at which one end is close and other end is open.

Stationary waves in closed organ pipe = Let

an close organ pipe of length l at the end of close organ pipe, there is

minimum vibration so these will mode.

At the open end of organ pipe particles vibration maximum, so those will be Antinodes



⇒ for first condition =

$$l = \frac{d_1}{4}$$

$$\text{Then } d_1 = 4l$$

If speed of air is v , then frequency is n_1

$$n_1 = \frac{v}{d_1}$$

$$n_1 = \frac{v}{4l} \quad \text{--- (1)}$$

This is a fundamental frequency or first harmonic

⇒ for 2nd condition =

$$l = \frac{3d_2}{4}$$

$$d_2 = \frac{4l}{3}$$

If frequency is n_2

$$n_2 = \frac{v}{d_2}$$

$$n_2 = \frac{v}{4/3 l}$$

$$n_2 = 3 \left(\frac{v}{4l} \right)$$

$$n_2 = 3n_1$$

→ It is 3rd Harmonic or first overtone

⇒ for 3rd condition

$$L = \frac{5d_3}{4}$$

$$d_3 = \frac{4L}{5}$$

frequency is n_3

$$n_3 = \frac{v}{d_3}$$

$$n_3 = \frac{v}{4L/5}$$

$$n_3 = 5 \left(\frac{v}{4L} \right)$$

$$n_3 = 5n_1$$

It is 5th Harmonic 2nd overtone

→ Ratio of frequency in close organ pipe

$$n_1 : n_2 : n_3 \dots$$

$$1 : 3 : 5 \dots$$

So In close organ pipe odd Harmonic can be produced.

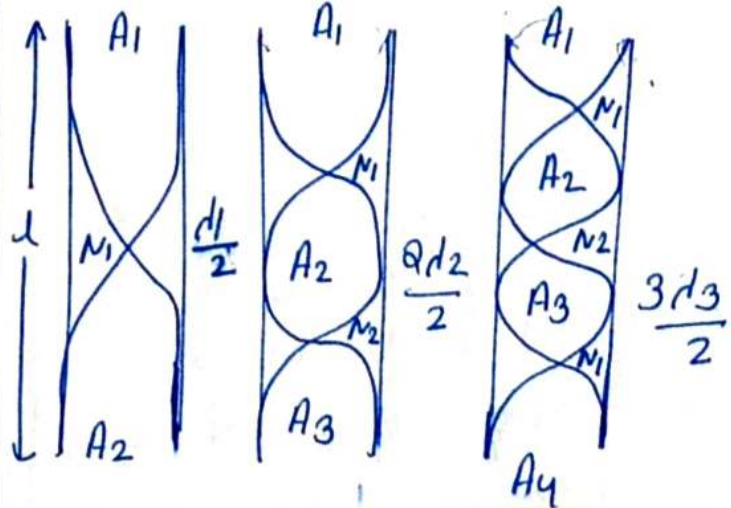
Stationary waves in open organ pipe =

→ The pipe which have Both ends open is called open organ pipe.

→ In open organ pipe two longitudinal waves travel in opposite direction.

→ Both ends are open so at Both ends there will be antinodes.

→ Let the length of organ pipe is L



→ In first case the distance B/w two antinodes is $\frac{d_1}{2}$

$$\Rightarrow L = \frac{d_1}{2}$$

$$\Rightarrow d_1 = 2L$$

If Velocity is v_1 then frequency is $n_1 = \frac{v}{d_1}$

$$n_1 = \frac{v}{2L}$$

It is a fundamental tone of first Harmonic

⇒ In 2nd case the distance B/w two antinodes is =

$$L = \frac{2d_2}{2}$$

$$d_2 = \frac{2L}{2}$$

frequency is n_2

$$n_2 = \frac{v}{d_2}$$

$$n_2 = \frac{v}{2L/2}$$

$$n_2 = 2 \left(\frac{v}{2L} \right)$$

$$n_2 = 2n_1$$

2nd Harmonic

⇒ In 3rd case the distance is

$$L = \frac{3d_3}{2}$$

$$n_3 = \frac{21}{3}$$

Frequency n_3

$$n_3 = \frac{v}{\lambda_3}$$

$$n_3 = \frac{v}{2\lambda/3}$$

$$n_3 = 3\left(\frac{v}{2\lambda}\right)$$

$n_3 = 3n_1$ 3rd Harmonic
or
2nd overtone

→ The ratio of frequency in open organ pipe

$$n_1 : n_2 : n_3 \dots$$

$$1 : 2 : 3 \dots$$

Beats = The regular rise and fall in the intensity of sound at constant time intervals at the same place is called beats.

Formation of Stationary waves in Air columns =

→ The minimum frequency of vibration of an object is called fundamental frequency

→ The tones that have frequencies more than the fundamental notes are called overtones. Like the tone of first high frequency after the fundamental note

is called first overtone.

organ pipe = The pipe in which air filled and by vibrating which sound waves can be produced is called organ pipe.

The diameter of organ pipe is very small compared to wavelength. But it is big enough so that the waves can easily come and go and so that the effect of viscosity of air is negligible, the walls should be rigid.

→ The pipe that has one end closed is called closed organ pipe.

→ The pipe that has both the ends open is called open organ pipe.

The principle of Superposition of wave =

$$y = y_1 + y_2 + \dots + y_n$$

Result of Superposition =

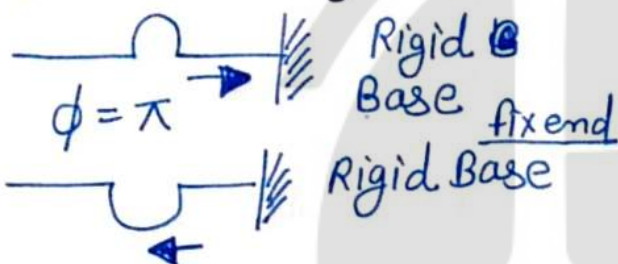
① Interference = Two wave of same f , same direction.

② Beats = Two sound waves of same frequency propagate in same direction

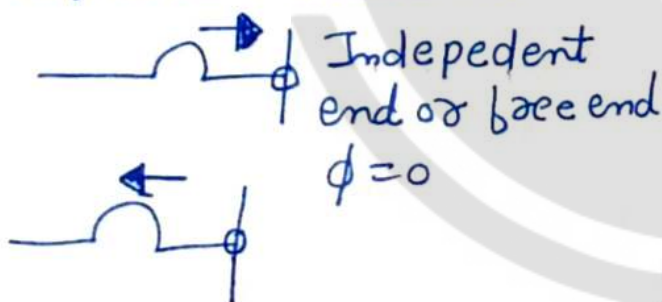
and Superimposed then this situation is called Beats.

⑦ Standing wave = when Two sound waves of same Amplitude and same frequency propagate in o/p direction along a simple line standing waves produced.

Reflection of waves =

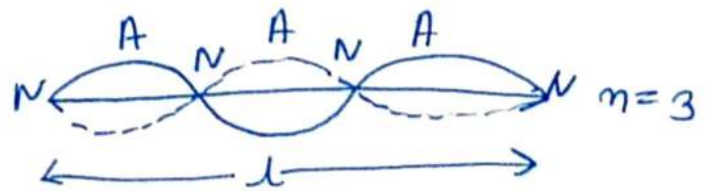
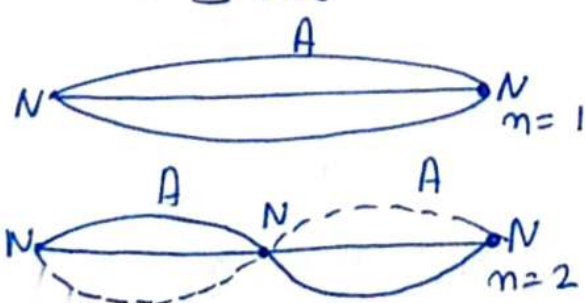


→ Incident disturbance and reflected disturbance



Standing waves in stretched string =

$$v = \sqrt{T/\mu}$$



⑧ minimum frequency sound
fundamental sound =

$$\Rightarrow l = \frac{\lambda}{2} \Rightarrow \lambda = 2l$$

$\Rightarrow$ fundamental frequency

$$n_1 = \frac{v}{\lambda} = \frac{\sqrt{T/\mu}}{\lambda}$$

$$n_1 = \frac{1}{2l} \sqrt{T/\mu}$$

⑨ 2nd Harmonic

$$\lambda = \frac{2l}{2} \Rightarrow \lambda_2 = \frac{2l}{2}$$

$$n_2 = \frac{v}{\lambda_2} = \frac{\sqrt{T/\mu}}{2l/2}$$

$$n_2 = \frac{2}{2l} \sqrt{T/\mu}$$

$n_2 = 2n_1$
1st overtone

$$n_1 : n_2 : n_3$$

$$1 : 2 : 3$$

Laws of Transverse vibration of a string =

① $n \propto \frac{1}{l}$ ② $n \propto \sqrt{T}$ ③ $n \propto \frac{1}{\sqrt{\mu}}$

standing wave in air column =

